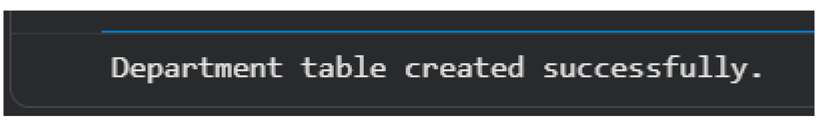
Activity using SQLite

1. Create the Department Table (Parent Table) with attributes Department_ID INTEGER PRIMARY KEY, Department_Name TEXT

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS Department (
    Department_ID INTEGER PRIMARY KEY,
    Department_Name TEXT
)
""")
conn.commit()
print("Department table created successfully.")
```



2. Create the Student Table (Child Table) with attributes: Student_ID INTEGER PRIMARY KEY, Name TEXT, Age INTEGER, Department_ID INTEGER, CGPA REAL, FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)

```
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
```

```
CREATE TABLE IF NOT EXISTS Student (
    Student_ID INTEGER PRIMARY KEY,
    Name TEXT,
    Age INTEGER,
    Department_ID INTEGER,
    CGPA REAL,
    FOREIGN KEY (Department_ID) REFERENCES
Department(Department_ID)
)
""")
conn.commit()
print("Student table created successfully.")
```

```
Student table created successfully.
```

Insert Sample Data

```
import sqlite3

conn=sqlite3.connect("college.db")
cursor=conn.cursor()

cursor.executemany("INSERT INTO Department VALUES (?,?)",[
(101,'CSE'),(102,'ECE'),(103,'EEE'),(104,'IT')])

cursor.executemany("INSERT INTO Student VALUES (?,?,?,?,?)",[
(1,'Rahul',20,101,8.9),
(2,'Priya',21,102,7.8),
(3,'Anjali',20,101,9.2),
(4,'Karthik',22,103,8.1),
(5,'Meena',21,104,8.7)])

conn.commit()
```

```
••• Data inserted successfully.
```

3. Display students from CSE department

```
cursor.execute("""
SELECT Student.*
FROM Student
JOIN Department
ON Student.Department_ID=Department.Department_ID
WHERE Department.Department_Name='CSE'
""")
print(cursor.fetchall())
```

```
Students from CSE Department
(1, 'Rahul', 20, 101, 8.9)
(3, 'Anjali', 20, 101, 9.2)
```

4. Count students in each department

```
cursor.execute("""
SELECT Department.Department_Name,
COUNT(Student.Student_ID)
FROM Department
LEFT JOIN Student
ON Department.Department_ID=Student.Department_ID
GROUP BY Department.Department_Name
""")
print(cursor.fetchall())
```

```
••• Student Count in Each Department
('CSE', 2)
('ECE', 1)
('EEE', 1)
('IT', 1)
```

5. Update the foreign key (change the department of a student)

```
cursor.execute("""
UPDATE Student
SET Department_ID=102
WHERE Student_ID=1
""")
```

conn.commit()

```
•••  Department updated successfully.
     (1, 'Rahul', 20, 102, 8.9)
     (2, 'Priya', 21, 102, 7.8)
     (3, 'Anjali', 20, 101, 9.2)
     (4, 'Karthik', 22, 103, 8.1)
     (5, 'Meena', 21, 104, 8.7)
```

6. Delete a department

```
cursor.execute("""
UPDATE Student
SET Department_ID=101
WHERE Department_ID=103
""")

cursor.execute("""
DELETE FROM Department
WHERE Department_ID=103
""")
conn.commit()
```

```
Department deleted successfully.
(101, 'CSE')
(102, 'ECE')
(104, 'IT')
```

7. Display the student's name, student's department and CGPA if the student's CGPA is greater than 8

```
cursor.execute("""
SELECT Student.Name,
Department.Department_Name,
Student.CGPA
FROM Student
JOIN Department
ON Student.Department_ID=Department.Department_ID
```

```
WHERE Student.CGPA>8
""")
print(cursor.fetchall())
```

```
...  Students with CGPA Greater Than 8
     ('Rahul', 'ECE', 8.9)
     ('Anjali', 'CSE', 9.2)
     ('Karthik', 'CSE', 8.1)
     ('Meena', 'IT', 8.7)
```